

1.

Find the output of the code:

```
#include <stdio.h>
int multiply(int a, int b)
{
    if (b == 0)
        return 0;
    return a + multiply(a, b - 1);
}
int cube(int n)
{
    if (n == 0)
        return 0;
    return multiply(n * n, n);
}
int main(void)
{
    int n = 3;
    printf("Cube of %d is %d\n", n, cube(n));
    return 0;
}
```

- A. Cube of 3 is 3
- B. Cube of 3 is 9
- C. Cube of 3 is 27
- D. Infinite recursion error

Answer: C

2.

Find the output of the code:

```
#include <stdio.h>
int fun(int x)
{
    static int s = 5;
    if (x > 0)
    {
        s += x;
        return fun(x - 1);
    }
    return s;
}
int main(void)
{
    printf("%d", fun(3));
    return 0;
}
```

- A. 5
- B. 6
- C. 11
- D. 14

Answer: C

3.

Find the output of the code:

```
#include <stdio.h>
void fun(int x);
int main(void)
{
    fun(3);    return 0;
}
```

```
void fun(int x)
{
    if (x > 0)
    {
        fun(x - 1);
        fun(x - 2);
        printf("%d ", x);
    }
}
```

- A. 1 2 3
- B. 1 2 2 3
- C. 1 2 1 3
- D. Too many outputs to predict

Answer: C

4.

What does this function compute?

```
int fun(int x, int y)
{
    if (y == 0)
        return 1;
    return x * fun(x, y - 1);
}
```

- A. x^y
- B. y^x
- C. $x+y$
- D. $x-y$

Answer: A

5.

Find the output of the code:

```
#include <stdio.h>
int fun(int x)
{
    if (x <= 0)
        return 0;
    else
        return x + fun(x - 2);
}
int main(void)
{
    printf("%d", fun(5));
    return 0;
}
```

- A. 9
- B. 8
- C. 10
- D. 12

Answer: A

6.

Find the output of the code:

```
#include <stdio.h>
int fun(int x)
{
    if (x < 10)
        return x;
    return fun(x / 10) + (x % 10);
}
int main(void)
{
    printf("%d", fun(123));
    return 0;
}
```

- A. 123
- B. 6
- C. 5
- D. 3

Answer: B